

ICE NAVIGATION

Welcome to study Maritime English: Ice Navigation. This module is divided into two parts. The first one is called *Ice navigation in theory*. It explains the main aspects of ice navigation in short texts based on research reports. In this part the exercises mainly test comprehension of the texts. The other part is called *Ice navigation in practice*. It consists of dialogues and exercises on these dialogues.

It must be noted that the dialogues are imaginary although real ship names and photos are used in the exercises. The traffic situations described in the dialogues have been created for the purposes of the MarEng Project.

Ice navigation, Unit 1

The Baltic Sea freezes annually

The two most heavily marine operated areas in the world where seasonal sea ice plays an important role in navigation are the Gulf of St. Lawrence in Canada and the Baltic Sea in Europe. In the Baltic Sea approximately 40 percent of the total amount of cargo turnover, about 700 million tons, occurs during winter months. The Baltic Sea freezes annually and in some parts the ice season lasts up to 7 months, from November to May. For example, in the Gulf of Finland the average length of the ice season is 120 days outside St. Petersburg, and 30 days at the entrance of the gulf.

The ice conditions are mostly affected by two factors: the number of sub-zero days (frost sum) and the prevailing winds. The sum of sub-zero degree-days controls the ice growth and the amount of ice. The prevailing winds control the drifting and ridging of an ice field.

Forming of an ice cover

An ice cover starts to form on water when the surface temperature reaches freezing point. Fresh water freezes at 0°C and in sea water the freezing point decreases with increasing salinity. Thus freezing point in ocean water is about -1.8°C, but in the brackish water of the Baltic Sea it is about -0.4°C.

In the Gulf of Finland ice thickness is greatest in the eastern parts of the gulf and is about 50 cm in an average winter. The biggest obstacles to winter navigation are ridges which are normally thicker than the level ice and are difficult to penetrate. Channels with thick side ridges and thick brash ice in the middle are formed when the ice cover in the fairway is repeatedly broken and refrozen. The side ridges make passing of other vessels very difficult. The keel heights of the ridges are normally a lot bigger than the sail heights. The side ridges may grow several metres thick and the brash ice layer in between may become up to one meter thick. Ridges also form when winds push ice together.

Why are ice classes needed?

The Ship Classification Societies' "ice class" has a fundamental basis on the safety of the ship hull and the essential propulsion machinery. The class defines sufficient installed power for safe operation in ice covered waters. The classification also defines certain hull structure against certain level ice, which in the Baltic Sea conditions is defined using the first-year ice definition. The classification also defines the requirements for the propeller shaft as minimum power for maintaining ship speed in a re-frozen (covered by e.g. brash ice) fairway navigation channel.

When the ice conditions become difficult, traffic restrictions are imposed. The restrictions pertain to the availability of icebreaker assistance. Some of the restrictions are about safety independent of assistance standards; some are caused by the availability of icebreaker services. The traffic restrictions are based on Ice Class Rules.

In the Baltic Sea area ice conditions are monitored on a daily basis. The Finnish Ice Service of the Institute of Marine Research issues ice charts and ice reports and produces ice drift forecasts. The daily ice chart and ice report include a description of current ice conditions and information about the icebreakers' operational areas.

This text has been adapted from the following sources:

Hänninen, Saara & Rytkönen, Jorma 2004: *Oil transportation and terminal development in the Gulf of Finland*. VTT Publications 547. VTT Technical Research Centre of Finland, Espoo. Pages 25, 28.



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Nyman, Tapio & Rytönen, Jorma 2004: The improvement of winter navigation with risk-based approaches. In *Ice Day. Basics of Winter Navigation in the Baltic Sea 11th and 12th of February, 2004*. Conference Report. C 4 / 2004. The Centre for Maritime Studies, University of Turku. Turku. Pages 220-258.

Seinä, Ari 2004: The different forms of ice. In Juhani Vainio (ed.) *Ice Day. Basics of Winter Navigation in the Baltic Sea 11th and 12th of February, 2004*. Conference Report. C 4 / 2004. The Centre for Maritime Studies, University of Turku. Turku. Pages 13, 16.

Ice navigation, Unit 2

Captain's checklist before entering ice-covered waters

It is always important to plan the voyage carefully for the safety and efficiency of navigation. Preplanning is especially important in winter time. The following checklist helps you to prepare properly for the voyage through ice-covered waters:

1. Start listening to the daily ice reports well in advance, and, if possible, order an ice chart.
2. Check that your VHF radio is operative, and find out the channel used by the icebreaker operating in the area.
3. Drain all water from the pipes on deck and empty containers of any liquids in case there is a danger of freezing.
4. Pump out all water from ballast tanks above the water-line if they are filled to the top in order to avoid freezing.
5. Protect the anchor windlass and the mooring hawser reels with a suitable tarpaulin before entering freezing conditions.
6. Keep the pilot ladder in a sheltered place so that it is not covered by ice when needed.
7. Test the searchlights.
8. Move the anchors astern or place them onto deck so that they may not come into contact with the icebreaker's towing notch. This has to be done in advance to prevent delays in assistance.
9. Check that the propeller and the rudder blade are deep enough below the waterline to ensure efficient operations in ice and to avoid air leaks to the propeller or

the rudder. If air is drawn either to the propeller or the rudder their efficiency will suffer considerably. Also, if the rudder and the propeller are too close to the waterline they will be much more vulnerable in icy conditions.

10. Check that main engine cooling water is available.

This text has been adapted from the following sources:

The Finnish Icebreaking Service, page 28. Available on the Internet:
www.fma.fi/toiminnot/talvimerenkulku/JM_isbr_icebr_2004-2005.pdf

Haapio, Antti 2002: *Navigation in Ice Infested Waters and Icebreaker Assistance*. Meriturva, Espoo. Page 19.

Ice navigation, Unit 3

Instructions to be followed in ice infested areas

- Avoid splitting thick and hard larger floes if you can go around them.
- Work with the ice and not against it.
- Respect the ice but do not be afraid of it.
- Do not rush - think first.
- Stay in open water within the natural leads and try to find the route of least resistance.
- Keep away from ridged ice.
- Keep the ship moving however slowly.
- Reverse the engine wisely and always set the rudder amidships when reversing.

This text has been adapted from the following source:

Haapio, Antti 2002: *Navigation in Ice Infested Waters and Icebreaker Assistance*. Revised edition. Meriturva, Espoo. Page 23.

Ice navigation, Unit 4

Convoy

When a vessel proceeds in a convoy, a careful watch shall be kept for signals coming from the icebreaker or another vessel in the convoy. During hours of darkness, icebreakers use a fixed blue all-around light at the top of the mast. If the icebreaker stops or slows speed unexpectedly, two rotating red warning lights, installed one upon the other, are lit. The master of the vessel assisted shall take all possible measures to stop his vessel as quickly as possible.

A vessel in convoy shall inform the icebreaker without delay if she stops or slows her speed substantially. If the vessel stops due to the ice condition, the searchlight must be switched off for as long as the vessel remains stationary.

This text has been adapted from the following source:

The Finnish Icebreaking Service, page 23. Available on the Internet:
www.fma.fi/toiminnot/talvimerenkulku/JM_isbr_icebr_2004-2005.pdf

Ice navigation, Unit 5

Towing

Notch towing

In difficult ice conditions, towing may be necessary. Three different towing methods are in use. Firstly, the vessel assisted by an icebreaker may be attached to the towing notch. This method called notch towing is the one normally used. The vessel's bow is brought into the towing notch of the icebreaker. The icebreaker hands over two steel-ropes which are fastened to the merchant vessel's bitts. (The bitts have been designed to withstand the stresses of towing.)

When notch towing is applied the hull of the towed vessel is acting as an active rudder of the icebreaker. When proceeding straight ahead the vessel should keep her masts in line with the icebreaker's masts.

When changing course, the helm has to be turned in a direction opposite to the one used normally, as the vessel's hull is acting as the rudder of the whole combination.

Towing a ship attached to the towing notch of the icebreaker is relatively safe. In the event that the icebreaker collides with packed ice and loses speed, the ship in tow does not damage the icebreaker, as it remains attached to the notch. The downside of this towing method is that the steering capabilities of the combination are poor.

Towing at a distance

Secondly, the vessel might be slightly distanced from the notch. The distance is so small that the stem of the assisted ship is able to move between the edges of the notch allowing small drift angles to develop when the icebreaker turns its rudders. This increases the turning capability of the tow. Sometimes a method where the stem is just outside the notch edges is used.

Thirdly, the vessel might be towed by means of a long cable. This method is used in open leads and also in order to save the very expensive protection "cushions" inside the notch. The heavier the vessel in tow the more likely it is that the cushions will be damaged.

The towing distance depends on the ship size, form of the fore head and the situation. Small vessels can normally be attached directly against the notch, while large ships are regularly towed with a distance of some tens of metres from the icebreaker.

When towing a ship at a small distance from the towing notch or with a long cable, the icebreaker must make sure that it can proceed at a steady speed. These two towing methods are only used in channels through fast ice, never in hard packed ice fields out in the sea. The advantage of these towing methods is preserving the capability of both the icebreaker and the vessel being towed to steer their course where required.

This text has been adapted from the following sources:

The Finnish Icebreaking Service, page 23-25. Available on the Internet: www.fma.fi/toiminnot/talvimerenkulku/JM_isbr_icebr_2004-2005.pdf



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Haapio, Antti 2002: *Navigation in Ice Infested Waters and Icebreaker Assistance*. Revised edition. Meriturva, Espoo. Page 44.

Uusiaho, Atso 2004: Navigation in ice - views of an icebreaker master in the Bay of Bothnia. In Juhani Vainio (ed.) *Ice Day. Basics of Winter Navigation in the Baltic Sea 11th and 12th of February, 2004*. Conference Report. C 4 / 2004. The Centre for Maritime Studies, University of Turku. Turku. Pages 145-150.

Ice navigation, Unit 6

On the vessel to be towed

Hoisting the towing cable and attaching it to the bollards of the assisted vessel should be carried out with the assistance of three persons. One person controls the anchor winch and two other persons work with the capstans hoisting the strop to level with the bollards. At least one of these persons should have enough experience on such attachment procedures. Heaving lines are needed on the assisted vessel to lift the messenger wires attached to the tow strop to hoist this up to the nocks. Both ends are lifted at the same time. The messenger wires should be guided by the bollard tops so that the strop can easily be pulled over to the bollard.

The icebreaker is in command during the towing operation. This means that the icebreaker gives all the commands and the vessel to be towed has to follow them without any delay. The crew on the assisted vessel should be ready to make fast or cast off the towing cable at any time. The propulsion machinery should be ready for rapid manoeuvres at all time according to instructions from the icebreaker. Also, it is the icebreaker who decides when the towing will be finished.

Letting go the tow

Letting go the tow is normally carried out by three men in the opposite manner to that of making fast the tow. The strop must not be let go in any circumstances, since it may well severely hit and damage the deck of the icebreaker when it slides through a bend. There are normally at least two persons on the deck of the icebreaker. Additionally, it is normal that the ice-breaker has to be manoeuvred during this process. There is a danger of getting the wires into the propellers of the ice-breaker.

The icebreaker slacks the cable so that the block connecting the cable to the strop is hanging straight down. The messenger wires or ropes are set on the winching drums of the windlass and the strop eyes pulled loose from the bollards. When this has been completed the wires are slowly lowered to the deck of the icebreaker. Note that the messengers are lowered using the heaving lines.

This text has been adapted from the following sources:

The Finnish Icebreaking Service, page 23 Available on the Internet: www.fma.fi/toiminnot/talvimerenkulku/JM_isbr_icebr_2004-2005.pdf

Haapio, Antti 2002: *Navigation in Ice Infested Waters and Icebreaker Assistance*. Revised edition. Meriturva, Espoo. Pages 43, 50-51.

Uusiaho, Atso 2004: Navigation in ice - views of an icebreaker master in the Bay of Bothnia. In Juhani Vainio (ed.) *Ice Day. Basics of Winter Navigation in the Baltic Sea 11th and 12th of February, 2004*. Conference Report. C 4 / 2004. The Centre for Maritime Studies, University of Turku. Turku. Pages 145-150.

Ice navigation, Unit 7

Icing

Icing is quite a common phenomenon near the ice edge during low temperatures and rough sea. Icing takes place when the air temperature is below 0 degrees Celsius but the sea is not covered by ice. Icing is caused by the lifting of spray into the relative wind by the ship bow. The spray is then super-cooled and carried over the ship superstructure to freeze on bulkheads, decks, and rigging. The water splashing on the deck freezes to the superstructures of the ship. The weight of accumulated ice can be significant. It can raise the centre of the gravity of the ship and deteriorate the ship's stability. Consequently, the ship may even capsize, if ice is not removed.

In case of ice build up on the deck structures appropriate tools should be available. Detaching the ice cover is a hard job. Hammers, even sledge hammers and hardwood clubs can be used but wisely. When hitting with a sledge hammer on a thick layer of ice the load on the metal sheet spreads rather evenly into the surrounding surface. In case the ice layer is thin the blow concentrates onto a very

small area and may force a dent into the plating. In such cases it is better to use heavy wooden clubs or even better leave it as it is.

This text has been adapted from the following sources:

Haapio, Antti 2002: *Navigation in Ice Infested Waters and Icebreaker Assistance*. Revised edition. Meriturva, Espoo. Page 20, 75.

Nyman, Tapio & Rytönen, Jorma 2004: The improvement of winter navigation with risk-based approaches. In Juhani Vainio (ed.) *Ice Day. Basics of Winter Navigation in the Baltic Sea 11th and 12th of February, 2004*. Conference Report. C 4 / 2004. The Centre for Maritime Studies, University of Turku. Turku. Pages 220-258.

Ice navigation, Unit 8

What's the damage?

The ice season brings with it characteristic accident types. Ice can cause different kinds of damage. They vary from main engine malfunction due to heavy ice conditions to paint finish degradation due to abrasion by ice pieces. In most cases the damage is small and limited to shell plating damage between a few frames at most.

There are a few typical situations which can cause damage to vessels navigating through ice or in an ice channel. The vessel may not be able to give way due to ice and may collide with another vessel. Also, vessels navigating in a compressive ice field might get stuck and sustain ice damage to the hull. Compressive ice induces high local loads that exceed the strength requirements of plating and framing. A vessel stuck in the ice may also drift with the moving ice field and may eventually run aground.

When ships are moving in an ice channel in succession with a short lead, the first ship may get stuck in the ice and the next one may collide with it. Reversing in ice may damage aft ship areas, especially the rudder and the propeller. In order to avoid getting stuck in ice, it is important that the propeller maintains its rotation. Usually the propeller ice loads are quite short-term and the so-called ice torque does not stop the propeller.

This text has been adapted from the following sources:

Hänninen, Samuli: Incidents and accidents in winter navigation in the Baltic Sea, winter 2002-2003. Available on the Internet:

<http://www.fma.fi/toiminnot/talvimerenkulku/pdf/No54.pdf> Pages 8-10.

Nyman, Tapio & Rytönen, Jorma 2004: The improvement of winter navigation with risk-based approaches. In *Ice Day. Basics of Winter Navigation in the Baltic Sea 11th and 12th of February, 2004*. Conference Report. C 4 / 2004. The Centre for Maritime Studies, University of Turku. Turku. Pages 220-258.



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Ice navigation, Dialogue 1

In a convoy

Situation description:

The M/S Marina is in the Gulf of Finland making her way towards Kotka, her port of destination. She has reached the longitude of Hanko (023 degrees East) and calls Helsinki VTS (Vessel Traffic Service). She wants to know the name of the coordinating icebreaker. According to the instructions given in the daily ice report, vessels bound for Finnish ports and requiring icebreaker assistance shall contact the icebreaker well in advance before entering ice-covered waters. The operator at the VTS centre saves the information given by the Marina to the knowledge base. The icebreakers in the Gulf of Finland can get the information from the knowledge base and can follow the passage of the vessel. When the Marina is closer to the nearest icebreaker, the navigating Deck Officer of the icebreaker gives her a Way Point and starts her way to meet the Marina. The icebreaker will take care of the Marina and assist her if needed.

Marina: Helsinki VTS, Helsinki VTS, Helsinki VTS, this is Marina, Marina, Marina, call sign MWYA3.

VTS: Marina, Helsinki VTS.

Marina: Helsinki VTS, Marina. We are passing Hanko, bound for Kotka. What is the first icebreaker?

VTS: Marina, Helsinki VTS. The first icebreaker is Kontio. At the moment she is close to Helsinki lighthouse. Kontio's working VHF channel is 08.

Marina: Okay. Thank you. The first icebreaker is Kontio, close to Helsinki lighthouse, working channel 08.

Marina gets closer to the icebreaker Kontio and calls her on channel 08.

Marina: Ice-breaker Kontio, Kontio, Kontio, this is Marina, Marina, Marina, call sign MWYA3.

Kontio: Marina, this is Kontio, good evening.

Marina: Kontio, Marina. My position is latitude 59 degrees and 49 minutes North, longitude 024 degrees and 18 minutes East and we are bound

for Kotka. Is the ice situation very bad there? Are you going to assist us?

Kontio: Marina, Kontio. We are close to Helsinki lighthouse. We are going westward with three other westbound ships towards south of Porkkala. After that we are coming to assist you. Stand by on channel 08. Proceed to the position 10 miles south of Helsinki lighthouse and wait for further instructions.

Marina: Kontio, Marina. Well understood. You are close to Helsinki lighthouse, going west with 3 ships towards south of Porkkala. After that you are coming to assist us. I will stand by on channel 08.

The icebreaker Kontio finishes assisting the three westbound vessels, leaves them in the fairway south of Porkkala, makes a turn and starts heading back eastwards to assist the Marina. After about 3 hours Kontio calls Marina on channel 08.

Kontio: Marina, Marina, Marina this is icebreaker Kontio, Kontio, Kontio on channel 08. Good evening, again.

Marina: Kontio, Marina. Good evening.

Kontio: Marina, Kontio. We are about one mile behind you here. We are going to pass you on your port side. Keep your heading and keep full ahead all the time.

Marina: Kontio, Marina. Okay, you are one mile behind me, you will pass me on my port side. We will keep full ahead on our current heading.

Kontio passes Marina and turns ahead of her.

Kontio: Marina, this is Kontio.

Marina: Kontio, Marina.

Kontio: Marina, Kontio. Please follow us now. Keep full ahead all the time. We will take care of the distance.

Marina: Kontio, Marina. Okay, we will keep full ahead all the time and you will take care of the distance.

Marina is following the icebreaker Kontio in an ice channel. While they proceed towards Lighthouse Helsinki Kontio takes two other ships in convoy. Five miles before meeting the first of these two vessels, Kontio calls her on the VHF channel 08. The icebreaker organizes the convoy, informs the vessels in the convoy about her plans and the voyage ahead. She also instructs the vessels to keep continuous listening watch on the working channel.

Kontio: Aila, Aila, Aila this is Kontio, Kontio, Kontio.

Aila: Kontio, Aila.

Kontio: Aila, Kontio. We are five miles west of you assisting Marina eastwards and we are going to take you with the convoy. Ship called Marina already behind us is number one and you will be number 2 in the convoy. We are going to pass you on the starboard side. After passing you, follow Marina.

Aila: Kontio, Aila. Okay, well received. You are five miles west of me assisting Marina, passing on starboard. We will join the convoy after Marina.

Kontio: Laura, Laura, Laura this is Kontio, Kontio, Kontio.

Laura: Kontio, Laura.

Kontio: Laura, Kontio. We are 4 miles southwest of you. We are assisting Marina. We are going to take Aila with the convoy. After that we will take you with the convoy. Marina is number one and Aila is number two. Your place in the convoy is number three. We will pass you on your port side. After we have passed you, follow Aila.

Laura: Kontio, Laura. Well understood. You are 4 miles southwest of me assisting Marina. You will pass me on my port side. I will be number three in the convoy. I will join the convoy after Aila.

Ice navigation, Dialogue 2

Towing

Situation description:

The M/S Marina is in the Gulf of Finland making her way towards Kotka, her port of destination. The edge of the ice cover reaches as far as Porkkala. The icebreaker Kontio has escorted the Marina to waypoint 10. After that the Marina has proceeded in an ice channel without icebreaker assistance. While Marina proceeds in the traffic separation scheme the ice cover gets thicker. The wind is getting stronger as well. At 1830 Marina gets stuck in position latitude 59 degrees and 56 minutes North, longitude 025 degrees and 36 minutes East. According to the instructions a vessel stuck in the ice must notify the icebreaker of her position without any delay. Marina calls the icebreaker Apu on the VHF channel 08.

Marina: Icebreaker Apu, Apu, Apu. This is Marina, Marina , Marina.

Apu: Marina, Apu replies.

Marina: Apu, Marina. We are ice-bound in position latitude 59 degrees and 56 minutes North, longitude 025 degrees and 36 minutes East.

Apu: Marina, Apu. Understood. Please prepare for towing. We are going to tow you in about 20 minutes.

Marina: Apu, Marina. Okay. We will prepare for towing. You are going to tow us in about 20 minutes.

Apu: Marina, Apu. Keep full ahead while preparing; we will tell you when to slow down.

Marina: Apu, Marina. Well received. We will keep full ahead.

Marina: Apu, this is Marina.

Apu: Marina, Apu.

Marina: Apu, Marina. I am ready for towing. My crew is on the forecastle head. We are ready to receive the towing lines.

Apu: Marina, Apu. Okay. Stop your speed.

Marina: Apu, Marina. Okay. We will stop our speed immediately.

Marina: Apu, Marina.

Apu: Marina, Apu.

Marina: Apu, Marina. I have stopped now. I am not moving.

Apu: Marina, Apu. Okay. You are not moving.

Apu reverses in front of Marina's bow.

Apu: Marina, Apu.

Marina: Apu, Marina.

Apu: Marina, Apu. Keep slow ahead.

Marina: Apu, Marina. Slow ahead.

Apu: Marina, Apu. We are going to give you two lines.

Marina: Apu, Marina. You are going to give us two lines. Do we need a heaving line?

Apu: Marina, Apu. Yes, you need a heaving line.

One of Marina's Able Body seamen casts a heaving line and Marina gets two lines. They are fastened to Marina's starboard and portside bollards on her forecastle head. An A.B. shows with his hands that the lines are made well fast.

Marina: Apu, Marina. Towing lines are made fast.

Apu: Marina, Apu. Towing lines fast. Ask your crew out of the forecastle. We are starting to move. Make half ahead. Steer with us and keep our masts in the same line.

Marina: Apu, Marina. Crew is clear of the forecastle. We are starting to move. We will make half ahead and steer with you. We keep our masts in the same line as yours.

Apu: Marina, this is Apu.

Marina: Apu, Marina.

Apu: Marina, Apu. We will finish towing in about 20 minutes.

Marina: Apu, Marina. Okay. Towing will be finished in 20 minutes.

Apu: Marina, Apu. Dead slow ahead.

Marina: Apu, Marina. Dead slow ahead.



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Apu: Marina, Apu. Keep slow ahead and keep our masts in the same line. Let go towing lines now.

Marina: Apu, Marina. Slow ahead, masts in the line. We are letting go towing lines.

Marina: Apu, Marina. We have let go towing lines now, the bow is clear.

Apu: Marina, Apu. Keep full ahead and follow us about 9 miles more.

Marina: Apu, Marina. Full ahead. We will follow you about 9 miles more.

Apu: Marina, Apu.

Marina: Apu, Marina.

Apu: Marina, Apu. We will finish assistance in about 10 minutes. You must proceed northeastwards alone. Are you ready to get the waypoints?

Marina: Apu, Marina. Understood. Assistance will be finished in 10 minutes. We are ready to take the waypoints.

Apu gives now several waypoints to Marina.

Apu: Marina, Apu. I will now give the waypoints to you. The next waypoint is 59 degrees 59 minutes North 026 degrees 00 minutes East. We will leave you in that position and you must proceed via waypoints on your own. Next icebreaker outside Orrengrund is "Voima". Her working channel is 06.

Marina: Apu, Marina. The next waypoint is 59 degrees 59 minutes North 026 degrees 00 minutes East. You will leave us in that position and I will continue alone via waypoints. Next Ib is Voima and if we need help again I will call Voima on channel 06.

Apu: Marina, Apu. I am finishing assistance now. I am turning hard to starboard.

Marina: Apu, Marina. Okay. You are finishing assistance now and turning to starboard.

Marina: Apu, Marina. Thank you very much for your assistance. Have a good watch.

Apu: Marina, Apu. Thank you Marina and good voyage.



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